Entanglement, Geometry, and the Limits of Quantum Observability

A Phenomenological Framework Coupling Quantum Entropy to Spacetime Curvature, with Two Independent Quantum Information Results

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Abstract

We present a connected three-paper research program at the boundary of quantum information, thermodynamics, and gravitational physics. **Paper A** proposes and analyzes a phenomenological modification of Einstein's field equations in which entanglement entropy density acts as a gravitational source term: $G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} + \tilde{\kappa}(k_{\mathrm{B}} \ln 2)^{-1} S_{\mathrm{ent}}\, g_{\mu\nu}$. Starting from Jacobson's thermodynamic derivation of general relativity, we estimate the coupling constant $\tilde{\kappa} = -1/4$ under an explicitly stated holographic regulator hypothesis, define an environmental screening factor α_{screen} that suppresses the coupling in decohered systems, and establish a five-condition falsification criterion that is achievable with near-term quantum hardware. We are explicit throughout about the extrapolation boundary: Jacobson's derivation applies rigorously to causal horizons, and this paper's extension to laboratory-scale entanglement is a physically motivated hypothesis, not a derivation. **Paper B** provides an independent, rigorously derived result in quantum information: the 40–50% entropy plateau observed in NISQ tomography is an epistemic artifact of undersampling, not a signature of physical decoherence. We prove this through a constraint manifold formalism, derive the shot-scaling law $\nu \propto 2^{n/2}$ from the quantum Cramér–Rao bound, and give a bias formula $\mathbb{E}[\hat{S}] = S_{\mathrm{vN}} + (d-1)/(2\nu) + B_{\mathrm{SPAM}}$ with an explicit correction procedure. **Paper C** derives Landauer's principle as a theorem from five physical axioms (causal closure, unitary microdynamics, effective thermodynamics, physical memory, finite resources), producing an observer-independent statement with three falsifiable predictions. All three papers are independently submittable; their connections are explicitly labeled.

Keywords: entanglement entropy; gravitational coupling; thermodynamics of spacetime;

quantum tomography; constraint manifolds; Landauer's principle; NISQ; quantum Fisher information

arXiv categories: Paper A: gr-qc (cross-list: quant-ph); Paper B: quant-ph; Paper C: quant-ph (cross-list: cond-mat.stat-mech)

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Epistemic Status Map

This document uses four epistemic labels applied consistently throughout:

- **[ESTABLISHED]** — follows from standard quantum mechanics, general relativity, or information theory; cited to verified literature.
- **[HYPOTHESIS]** — physically motivated extension beyond the proven domain of established theory; the specific assumption is stated explicitly each time.
- **[ANALOGY]** — structural parallel between physical and non-physical systems; does not constitute a physical derivation.
- **[OPEN]** — a gap in the derivation that is flagged rather than silently smoothed over.

The core scientific integrity requirement of this program is the maintenance of a strict boundary between [ESTABLISHED] and [HYPOTHESIS]. A hypothesis can be valuable, interesting, and worth publishing. It becomes fraudulent only when it is mislabeled as established.

Paper A: Gravitational Coupling to Entanglement Entropy Density

1 Introduction and Motivation

The relationship between quantum information and gravitational geometry has undergone substantial development since Bekenstein (1972) and Hawking (1975) established that black hole entropy scales with horizon area. Jacobson's 1995 derivation showed that Einstein's field equations follow as a thermodynamic equation of state when entropy is assumed to scale with area at local Rindler horizons [1]. More recently, Verlinde proposed that even the Newtonian gravitational force emerges from the entropic tendency of matter to maximize the entropy of the screens surrounding it [2]. ER=EPR connects entanglement structure to spacetime connectivity at the Planck scale [3].

A natural question left open by this literature is: does laboratory-scale quantum entanglement — entanglement created in and maintained by experimental quantum systems far from any causal horizon — produce a measurable contribution to spacetime curvature? The horizon-based results established that information and geometry are coupled at the extreme end of the scale. This paper asks whether that coupling, appropriately weakened, persists in intermediate regimes.

We do not answer this question. We frame it precisely, derive a dimensionally consistent coupling ansatz from Jacobson's framework under an explicitly stated extrapolation hypothesis, estimate the coupling constant under that hypothesis, analyze the environmental suppression that makes the coupling undetectable in typical laboratory conditions, and specify the experimental conditions under which the hypothesis can be falsified or confirmed.

What this paper claims:

- A dimensionally consistent phenomenological modification of the Einstein field equations in which entanglement entropy density sources curvature (Sec. 2).
- An estimate $\tilde{\kappa} = -1/4$ under a holographic regulator hypothesis that is stated precisely and labeled as a hypothesis (Sec. 3).
- A screening analysis showing the coupling is suppressed by environmental decoherence, with an open problem on the size of that suppression (Sec. 5).
- A falsification criterion that is meaningful regardless of whether the signal magnitude is currently computable (Sec. 8).

What this paper does not claim:

- That entanglement sources curvature has been derived from first principles. The extrapolation from horizons to laboratory systems is a hypothesis.

- That a predicted signal magnitude is available. The signal size is currently indeterminate (Open Problem 10, Sec. 3.4).
- That current hardware can detect the effect. The proposed experiment is a near-term target, not a present capability.

2 The Modified Einstein Field Equation

2.1 Standard General Relativity [ESTABLISHED]

Einstein's field equations in geometric units with $c = 1$, restored to SI for dimensional clarity:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{A1.1}$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$ is the Einstein tensor (units: m^{-2}), $T_{\mu\nu}$ is the stress-energy tensor (units: $\mathrm{kg\,m^{-1}s^{-2}}$), and $G = 6.674 \times 10^{-11}\ \mathrm{m^3kg^{-1}s^{-2}}$. Metric signature $(-, +, +, +)$ throughout.

2.2 The Proposed Coupling [HYPOTHESIS]

We propose adding an entanglement entropy source term:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu} + \tilde{\kappa} \, \frac{c^4}{8\pi G \, k_{\mathrm{B}} \ln 2} \, S_{\mathrm{ent}} \, g_{\mu\nu} \right) \tag{A1.2}$$

where S_{ent} is the *entanglement entropy density* (bits/m^3; always a density, never a total entropy), $\tilde{\kappa}$ is a dimensionless coupling constant, and $k_{\mathrm{B}} \ln 2 = 9.570 \times 10^{-24}$ J/K converts bits to thermodynamic entropy units.

2.2.1 Dimensional Verification [ESTABLISHED]

The entanglement term must have units of $T_{\mu\nu}$, i.e., $\mathrm{kg\,m^{-1}s^{-2}}$. Checking:

$$\begin{aligned} \left[\frac{c^4}{8\pi G \, k_{\mathrm{B}} \ln 2} \, S_{\mathrm{ent}} \right] &= \frac{(\mathrm{m/s})^4}{\mathrm{m^3kg^{-1}s^{-2} \cdot J\,K^{-1}}} \cdot \frac{\mathrm{bits}}{\mathrm{m^3}} \\ &= \frac{\mathrm{m^4s^{-4}}}{\mathrm{m^3kg^{-1}s^{-2} \cdot kg\,m^2s^{-2}}} \cdot \mathrm{m^{-3}} \\ &= \mathrm{kg\,m^{-1}s^{-2}} \quad \checkmark \end{aligned} \tag{A1.3}$$

Dimensional consistency is satisfied. Note that dimensional consistency is a necessary condition for a physical equation, not a sufficient one.

2.2.2 The Repulsive Gravity Condition

For $\tilde{\kappa} < 0$ (as derived in Sec. 3) and $S_{\text{ent}} > 0$ (entropy density is non-negative by definition), the entanglement term contributes negative effective pressure. The active gravitational mass density becomes:

$$\rho_{\text{grav}} + \frac{3p_{\text{grav}}}{c^2} = \rho_m + \frac{3p}{c^2} + \frac{3\tilde{\kappa}c^2}{8\pi G\, k_{\text{B}} \ln 2}\, S_{\text{ent}} \tag{A1.4}$$

With $\tilde{\kappa} < 0$, the entanglement term *reduces* the active gravitational mass — producing a repulsive effect without exotic matter, negative mass, or modified geometry. The repulsion emerges from the quantum information structure of the state.

2.3 Ontological Constraints

This modification is designed to be minimal: the classical spacetime manifold and metric signature are unchanged; standard quantum mechanics governs matter fields; no new particles or exotic matter are introduced; Newton's constant G is unchanged; the Einstein tensor $G_{\mu\nu}$ retains its standard definition. The only addition is the source term on the right-hand side of (A1.2).

3 Estimating the Coupling Constant

3.1 Jacobson's Thermodynamic Foundation [ESTABLISHED]

Jacobson (1995) [1] derived Einstein's field equations from thermodynamics. The essential ingredients are:

- **Local Rindler horizon.** For any spacetime point, an observer with proper acceleration a perceives a local Rindler horizon and an Unruh temperature:

$$T_U = \frac{\hbar a}{2\pi c\, k_{\text{B}}} \tag{A2.1}$$

- **Bekenstein-Hawking entropy.** The entropy of a horizon area element dA is:

$$dS_{BH} = \frac{k_{\text{B}} c^3}{4G\hbar}\, dA \tag{A2.2}$$

- **Clausius relation.** Applying $\delta Q = T\, dS$ to the heat flux through the horizon recovers (A1.1) exactly.

This is a rigorous derivation. Gravity is, under this reading, the equation of state of spacetime entropy. Every step is established physics.

3.2 The Holographic Regulator Hypothesis

[HYPOTHESIS — the extrapolation step]

Holographic Regulator Hypothesis. We propose that an entanglement entropy contribution to the Clausius relation takes the form:

$$dS_{\text{ent}} = \frac{S_{\text{ent}}}{k_{\text{B}}} \cdot \frac{\ell_P \, dA}{4} \tag{A2.3}$$

where $\ell_P = \sqrt{\hbar G/c^3} = 1.616 \times 10^{-35}$ m is the Planck length and $\ell_P \, dA$ is the volume element one Planck length behind the horizon.

This treats one Planck length of depth as the effective thickness through which entanglement entropy contributes to horizon thermodynamics. The regulator is borrowed by analogy from holographic principles (area-law entropy at causal horizons). **It is assumed to extend to laboratory-scale entanglement volumes without any horizon present. This extension has no derivation.** Everything that follows is internally consistent conditional on this assumption.

3.3 Deriving $\tilde{\kappa} = -1/4$ [HYPOTHESIS → ESTIMATE]

The modified Clausius relation becomes:

$$\delta Q_{\text{eff}} = T_U \, dS_{BH} + T_U \, dS_{\text{ent}} = \delta Q_{BH} + \frac{\hbar a}{2\pi c \, k_{\text{B}}} \cdot \frac{S_{\text{ent}}}{k_{\text{B}}} \cdot \frac{dA}{4} \tag{A2.4}$$

Following Jacobson's identification procedure, the entanglement term contributes an effective stress-energy:

$$T^{\text{ent}}_{\mu\nu} = -\frac{c^4}{32\pi G} \, S_{\text{ent}} \, g_{\mu\nu} \tag{A2.5}$$

Comparing with the proposed form $T^{\text{ent}}_{\mu\nu} = \tilde{\kappa} \, \frac{c^4}{8\pi G \, k_{\text{B}} \ln 2} S_{\text{ent}} \, g_{\mu\nu}$:

$$\boxed{\tilde{\kappa} = -\frac{1}{4} \quad \text{[under holographic regulator hypothesis]}} \tag{A2.6}$$

This derivation is internally consistent. Its validity is entirely conditional on the holographic regulator hypothesis of Sec. 3.2, which is not derived. The value $\tilde{\kappa} = -1/4$ should therefore be treated as a *dimensionally and thermodynamically motivated estimate*, not a first-principles derivation.

3.4 Relation to the Planck Scale and Renormalization

[OPEN PROBLEM OP10]

The coupling prefactor $c^4/(8\pi G\, k_{\rm B} \ln 2) \approx 5 \times 10^{65}$ m^{-3} is not numerically innocent. It is large because c^4/G is the Planck energy density scale while $k_{\rm B} \ln 2$ is far below Planck energy. A naive application of (A1.2) with $\tilde{\kappa} = -1/4$ and laboratory-scale $S_{\rm ent}$ produces signals that are either impossibly large (for optimistic $\alpha_{\rm screen}$) or immeasurably small (for realistic $\alpha_{\rm screen}$ from spatial suppression models).

This mismatch has three candidate resolutions:

1. The Newtonian limit derivation from (A1.2) to an acceleration correction $\Delta a(R)$ requires an explicit source profile and may not be straightforward.
2. The coupling $\tilde{\kappa} = -1/4$ is a Planck-scale quantity that renormalizes to far smaller values at laboratory scales, analogously to how Newton's constant is $\sim 10^{38}$ times weaker than gauge couplings.
3. The framework may be non-perturbative at laboratory scales in a way that invalidates the Newtonian-limit comparison.

Consequence for this paper: The signal magnitude is indeterminate and is not quoted as a prediction. The falsification criterion of Sec. 8 is stated in terms of directly measurable bounds on $|\tilde{\kappa}|$, not in terms of a computed signal. The theoretical framework stands independently of whether the signal is currently computable.

4 The Extrapolation Boundary

4.1 Where Jacobson's Derivation Applies [ESTABLISHED]

Jacobson's result holds under three conditions that are all satisfied at causal horizons and none of which hold in a typical laboratory:

1. A causal horizon exists.
2. A well-defined Unruh temperature $T_U = \hbar a/(2\pi c k_{\rm B})$ exists (requires steady-state uniform acceleration).
3. Entanglement entropy scales with *area*, not volume (the area-law property of horizon entanglement).

A 10^6-atom atomic ensemble has no causal horizon, no well-defined Unruh temperature, and volume-law entanglement entropy (for a GHZ state, entropy scales with atom number, not surface area).

4.2 The Extrapolation Hypothesis, Precisely Stated

[HYPOTHESIS]

Extrapolation Hypothesis. The coupling between entanglement entropy density and spacetime curvature, which follows rigorously from area-law entropy at causal

horizons, extends to systems with volume-law entanglement entropy in the absence of causal horizons, regulated by the Planck-scale geometric structure that governs horizon area entropy.

This is a physical hypothesis. It cannot be derived from established physics. The scientific case for investigating it rests on four considerations: (1) holographic universality suggests area-law entropy at horizons is the extreme limit of a more general information-geometry coupling; (2) the Bose et al. proposal for gravity-mediated entanglement [4] demonstrates that gravity and quantum entanglement interact at laboratory scales beyond classical gravity; (3) the dimensional analysis of Sec. 2 shows the coupling *can* be written consistently; (4) the hypothesis is falsifiable by a realistic experiment.

4.3 Removed Language

The following formulations appeared in source drafts and are *removed* from this paper:

- × "First-principles derivation of κ" → ✓ "Estimate under holographic regulator hypothesis"
- × "We prove that entanglement entropy sources curvature" → ✓ "We propose and analyze the hypothesis that..."
- × "No horizon is required" → ✓ "We hypothesize no horizon is required; this is not established"

5 Environmental Screening

5.1 Physical Motivation [HYPOTHESIS]

The ideal coupling $\tilde{\kappa} = -1/4$ applies to a perfectly isolated, maximally coherent quantum system. Real systems are continuously coupled to their environments through photon scattering, phonon interactions, magnetic fluctuations, electromagnetic interference, and residual gas collisions. Each interaction transfers entanglement from within the system to system-environment correlations, removing it from the internal S_{ent} that sources curvature. The effective coupling is:

$$\tilde{\kappa}_{\text{eff}} = -\frac{1}{4}\,\alpha_{\text{screen}}, \quad \alpha_{\text{screen}} \in [0, 1] \tag{A4.1}$$

where $\alpha_{\text{screen}} = 1$ is perfect isolation and $\alpha_{\text{screen}} = 0$ is complete decoherence.

5.2 Phenomenological Formula and Its Limitations

[OPEN PROBLEM OP3]

A physically motivated ansatz:

$$\alpha_{\text{screen}} = \frac{1}{1 + (\Gamma \tau_D)^2} \cdot e^{-(R/\xi)^2} \tag{A4.2}$$

where Γ is the decoherence rate, $\tau_D = 1/\Gamma$ the decoherence time, R the spatial extent of the coherent region, and ξ the coherence length.

Critical limitation: The Gaussian spatial factor $e^{-(R/\xi)^2}$ produces $\alpha_{\text{screen}} \sim e^{-100} \approx 10^{-43}$ for a macroscopic ^{87}Rb ensemble with $R = 100\,\mu$m and $\xi = 10\,\mu$m — far outside the heuristic range $[10^{-4}, 10^{-2}]$ cited in earlier drafts of this program. That heuristic range applies to few-qubit systems ($R \sim \xi$, i.e., $R \sim 0.1$–1 mm), not to macroscopic atomic ensembles.

Status: The range $\alpha_{\text{screen}} \in [10^{-4}, 10^{-2}]$ was not derived from first principles in any version of this work. It should be treated as provisional. The derivation of α_{screen} from the Lindblad equation with explicit system parameters remains an open problem (OP3) that constitutes a graduate-level research project.

5.3 Consequence for the Experimental Target

Because the macroscopic ensemble regime may have $\alpha_{\text{screen}} \sim 10^{-43}$, the preferred near-term experimental target shifts to 50–100 qubit NISQ-scale systems where $R \sim \xi$, entanglement structure is programmable, and state verification is more direct.

6 Newtonian Limit and the Radial Correction

6.1 Enclosed-Source Formula [ESTABLISHED framework]

Assuming spherical symmetry and a modified Poisson equation for the entanglement-sourced gravitational potential Ψ:

$$\frac{1}{R^2}\frac{d}{dR}\left[R^2 \frac{d\Psi}{dR}\right] = \kappa\, \rho_{\text{ent}}(R) \tag{A5.1}$$

Defining the enclosed entanglement entropy: $M_{\text{ent}}(< R) = 4\pi \int_0^R \rho_{\text{ent}}(r)\, r^2\, dr$, the radial acceleration correction is:

$$\Delta a(R) = -\frac{d\Psi}{dR} = \frac{\kappa\, M_{\text{ent}}(< R)}{R^2} \tag{A5.2}$$

6.2 Scaling Behavior

For a compact (finite) source, $M_{\text{ent}}(< R)$ saturates at large R, so $\Delta a(R) \propto 1/R^2$ outside the source — the same scaling as Newtonian gravity. A $1/R$ correction, appearing in earlier source drafts, would require a source profile $\rho_{\text{ent}}(r) \propto 1/r^2$, which is physically unmotivated and is *not* adopted here.

6.3 Status of the Signal Prediction

Given OP10 (Sec. 3.4) and OP3 (Sec. 5), no specific numerical prediction for $|\Delta a|$ is made. The formula (A5.2) is structurally correct. The predicted magnitude is indeterminate pending resolution of the Planck-scale renormalization question and the derivation of α_{screen}.

7 Experimental Falsification Protocol

7.1 Concept and Target Regime

The core observable is a differential acceleration (or phase shift) between two systems prepared to be identical in all respects except their entanglement structure: one arm in a coherent, high-entanglement state; the other deliberately decohered or prepared in a low-entanglement reference state.

The near-term target is a **50–100 qubit NISQ-scale platform** where entanglement depth, geometry, and decoherence channels are explicitly controlled. This replaces the earlier $N = 10^6$ atom target, which is inappropriate given the α_{screen} analysis (Sec. 5).

7.2 Measurement Protocol

Branch A. Prepare a 50–100 qubit state with maximal or near-maximal multipartite entanglement.

Branch B. Prepare a matched reference state with the same qubit count, control sequence, and gross energy budget, but with entanglement intentionally suppressed.

Readout. Compare entanglement-conditioned phase, acceleration proxy, or equivalent interferometric observable across repeated trials.

Controls. Interleave randomized compilation, device-bias reversals where available, and explicit entanglement-witness measurements to verify branch contrast.

7.3 Key Systematic Concerns

Table 1: Systematic error considerations for the NISQ-scale experiment.

Systematic	**Why it matters**	**Mitigation**
Imperfect state preparation	Can mimic entanglement-dependent signal	Verify entanglement depth before/after
Decoherence drift over time	Changes branch contrast	Randomize branch order; short calibration cycles
Geometry-dependent coupling	Affects theory-to-signal mapping	Simulate with explicit source profile
Platform cross-talk	Creates false branch asymmetries	Symmetrize control pulses; null reference circuits
Ambiguous α_{screen} regime	Prevents trustworthy signal forecast	Treat signal as indeterminate; focus on upper bounds

8 Falsification Criterion

A theory is falsifiable only if specific experimental outcomes can rule it out. The single free parameter of this framework is $\tilde{\kappa}$. If $|\tilde{\kappa}|$ can always be made smaller than any experimental sensitivity by adjusting α_{screen} post-hoc, the framework becomes unfalsifiable. We close this escape hatch.

Falsification Criterion. The framework is *falsified for laboratory-scale relevance* if all five conditions below are simultaneously satisfied:

F1 (System size). Quantum coherent systems with $\geq 10^6$ genuinely entangled qubits are achieved (entanglement depth $\geq 10^3$, verified by entanglement witness, not just claimed by state preparation).

F2 (Sensitivity). Experimental pressure sensitivity $\Delta p < 10^{-6}$ Pa, corresponding to $\delta a < 10^{-12}$ m/s^2.

F3 (Statistics). At least 1000 independent experimental runs with statistically consistent null results.

F4 (Platform independence). Null results on ≥ 3 physically distinct platforms (e.g., trapped neutral atoms, superconducting qubits, optomechanical oscillators).

F5 (No anomalous signal). No stress-energy contribution beyond standard Lindblad decoherence models at F2 sensitivity after F3 statistics on F4 platforms.

Consequence: Satisfying F1–F5 establishes $|\tilde{\kappa}| < 10^{-15}$, ruling out laboratory-engineering relevance of the coupling for any foreseeable technology.

Satisfying F1–F5 does not rule out astrophysical or cosmological effects, nor a coupling smaller than 10^{-15}. What it rules out is the strong form of the hypothesis: that entanglement gravity is detectably relevant at laboratory scales within decades.

8.1 Detection Thresholds

Table 2: Interpretation of detection results.

Result	**Interpretation**
$\lvert\tilde{\kappa}\rvert \sim 10^{-4}$–$10^{-3}$	Strong confirmation; $\alpha_{\text{screen}} \in [10^{-3}, 10^{-2}]$; engineering implications
$\lvert\tilde{\kappa}\rvert \sim 10^{-5}$–$10^{-6}$	Weak confirmation; astrophysical implications
$\lvert\tilde{\kappa}\rvert \sim 10^{-8}$–$10^{-12}$	Marginal; requires revision of α_{screen} model
$\lvert\tilde{\kappa}\rvert < 10^{-15}$ (with F1–F5)	Falsification of laboratory relevance

8.2 Timeline Estimate

Current technology reaches $N \sim 10^4$–10^5 entangled atoms with $\delta a \sim 10^{-11}$ m/s^2 [5]. Achieving F1 (entanglement depth $\geq 10^3$ across 10^6 qubits) and F2 ($\delta a < 10^{-12}$ m/s^2) requires incremental improvements achievable within 5–8 years on existing experimental trajectories. No new physics is required for the experiment itself, only for the interpretation of a positive result.

Open Problems — Paper A

OP3 (Critical). Derivation of α_{screen} from the Lindblad equation with explicit parameter values for the proposed system. The heuristic range $[10^{-4}, 10^{-2}]$ is not derived; the spatial Gaussian model gives $\sim 10^{-43}$ for macroscopic ensembles. Months of graduate-level work.

OP8 (High). Explicit verification of Bianchi identity compatibility for the modified field equation (A1.2). Standard matter conservation $\nabla^\mu T_{\mu\nu} = 0$ plus the new S_{ent} source term must together satisfy $\nabla^\mu G_{\mu\nu} = 0$. Approximately one week of calculation.

OP10 (Critical, long-term). Resolution of whether the coupling $\tilde{\kappa} = -1/4$ is a Planck-scale bare value subject to renormalization at laboratory scales, and whether the Newtonian limit derivation is valid.

Citation note (to be verified before submission): The Bose et al. result cited in this paper as evidence for gravity-mediated entanglement is the 2017 theoretical proposal [4]. Any reference to a "Nature 623" experimental result must be traced to a specific paper before appearing in the final manuscript.

Paper B: Constraint Manifolds and the Limits of Quantum Observability

9 Introduction

A recurring observation in NISQ quantum computing experiments is that quantum state tomography on systems with $n \geq 10$ qubits consistently returns von Neumann entropy estimates of 40–50% of the theoretical maximum, regardless of how carefully the system is prepared. This plateau is often attributed to physical decoherence: the system, it is assumed, has been partially thermalized by coupling to its environment.

Central Claim (Paper B). For typical NISQ tomography budgets and qubit counts above 10, the 40–50% entropy plateau is entirely epistemic — an artifact of insufficient measurement, not a signature of physical decoherence. It persists for *any* physical state, coherent or mixed, until the shot count reaches $\nu \propto 2^{n/2}$.

This is a practical result with immediate implications: tomography results from underpowered experiments are being systematically misinterpreted, and the misinterpretation can be corrected by applying the bias formula derived here.

10 The Constraint Manifold Formalism

10.1 State Space Hierarchy

Definition 10.1 (Density Operator Space)**.** For an n-qubit system, $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$. The full space of density operators is:

$$\mathcal{D} = \left\{\rho \in \mathcal{L}(\mathcal{H}) \mid \rho = \rho^\dagger,\ \rho \geq 0,\ \mathrm{Tr}\rho = 1\right\} \tag{B1.1}$$

of real dimension $4^n - 1$.

Definition 10.2 (Constraint Manifold)**.** The *constraint manifold* is the subset of $\mathcal{D}$ consistent with all irreversible physical constraints:

$$\mathcal{C}_{\text{constr}} = \left\{\rho \in \mathcal{D} \mid \mathrm{Tr}(\hat{C}_i \rho) = c_i \ \ \forall i \in \mathcal{I}\right\} \tag{B1.2}$$

where $\hat{C}_i$ are Hermitian constraint operators (conserved quantities such as energy and particle number), c_i their fixed values, and $\mathcal{I}$ indexes irreversible constraints.

Definition 10.3 (Observable Subspace)**.** Given measurement history $r = \{(i_1, c_{i_1}), \ldots, (i_k, c_{i_k})\}$:

$$\mathcal{C}_{\text{obs}}(r) = \left\{\rho \in \mathcal{C}_{\text{constr}} \mid \mathrm{Tr}(\hat{C}_{i_j} \rho) = c_{i_j} \ \ \forall j \leq k\right\} \tag{B1.3}$$

As k increases, $\mathcal{C}_{\text{obs}}(r)$ *shrinks* — not because the system decoheres, but because we know more about it.

Proposition 10.4 (Dimension of the Observable Subspace)**.** *For* $\mathrm{rank}(C)$ *linearly independent accumulated constraints:*

$$\dim \mathcal{C}_{\mathrm{obs}}(r) = 4^n - \mathrm{rank}(C) - 1 \tag{B1.4}$$

Proof. $\mathcal{D}$ has dimension $4^n - 1$. Each linearly independent equality constraint $\mathrm{Tr}(\hat{C}_i \rho) = c_i$ removes exactly one real degree of freedom (restricts to a hyperplane in Hermitian operator space). With $\mathrm{rank}(C)$ independent constraints, dimension reduces by $\mathrm{rank}(C)$, giving $4^n - 1 - \mathrm{rank}(C)$. □

10.2 Why Undersampling Produces Entropy Bias

For a 15-qubit system ($n = 15$, $d = 2^{15} = 32{,}768$):

$$\dim \mathcal{D} = 4^{15} - 1 \approx 10^9 \tag{B1.5}$$

Standard NISQ tomography with $\nu = 1000$ shots yields $\mathrm{rank}(C) \approx 3000$, so:

$$\dim \mathcal{C}_{\mathrm{obs}}(r) \approx 10^9 - 3 \times 10^3 \approx 10^9 \tag{B1.6}$$

The observable subspace is essentially the entire density matrix space. Because the vast majority of $\mathcal{D}$ consists of high-entropy (nearly maximally mixed) states, any estimator derived from the measurement record is biased toward high entropy. This is a geometric fact about the space of density matrices, not a statement about the physical system.

11 Quantum Fisher Information and the Shot-Scaling Law

11.1 The Cramér–Rao Bound for Entropy Estimation

Theorem 11.1 (Quantum Cramér–Rao Bound)**.** *For any unbiased estimator* $\hat{S}$ *of the von Neumann entropy* $S_{\mathrm{vN}}(\rho)$ *based on* ν *independent shots:*

$$\mathrm{Var}[\hat{S}] \geq \frac{[\mathcal{F}_Q(\rho)]^{-1}}{\nu} \tag{B2.1}$$

where $\mathcal{F}_Q(\rho) = \mathrm{Tr}[\rho L^2]$ *is the quantum Fisher information and* L *is the symmetric logarithmic derivative.*

Remark 11.2. $\mathcal{F}_Q(\rho)$ is a functional of the quantum *state* ρ, not of the scalar $S_{\mathrm{vN}}(\rho)$. Two states with identical von Neumann entropy can have different $\mathcal{F}_Q$ (e.g., a thermal Gibbs state vs. a random pure state at the same entropy). The earlier notation $\mathcal{I}_Q^{-1}(S_{vN})$ appearing in source drafts is *incorrect* and is replaced throughout by $\mathcal{F}_Q(\rho)$.

11.2 The $2^{n/2}$ Scaling Law

Proposition 11.3 (Shot Scaling for Entropy Estimation)**.** *Near the maximally mixed state $\rho \approx \mathbf{I}/d$:*

$$\mathcal{F}_Q(\rho)\Big|_{\rho\approx\mathbf{I}/d} \sim 2^{1-n} \tag{B2.2}$$

so that achieving entropy precision ε requires:

$$\nu \gtrsim \frac{2^{n+1}}{\varepsilon^2} \propto 2^n \tag{B2.3}$$

For states at entropy $S_{\mathrm{vN}} = \alpha n$ (the plateau regime, $\alpha \approx 0.4$–0.5), the required shots for distinguishing from a pure state scale as:

$$\nu_{\text{plateau}} \approx \frac{c(\alpha) \cdot 2^{n/2}}{\varepsilon^2} \tag{B2.4}$$

This is the $\nu \propto 2^{n/2}$ ***scaling law****: the half-exponent arises because the entropy estimation problem is halfway in complexity between single-observable estimation ($O(1)$) and full tomography (4^n).*

Table 3: Required shot counts for entropy estimation ($\varepsilon = 0.1$ bits) versus typical NISQ budgets.

Qubits	**Hilbert dim d**	**Required ν**	**Typical NISQ**	**Gap**
5	32	≈ 560	10^3	$< 1\times$
10	1,024	$\approx 3{,}200$	10^3	$3\times$
15	32,768	$\approx 18{,}000$	10^3	$18\times$
20	1.05×10^6	$\approx 100{,}000$	10^3	$100\times$
25	3.3×10^7	$\approx 580{,}000$	10^3	$580\times$

The plateau begins at $n \approx 10$ precisely because that is where the required shot count first exceeds the typical NISQ budget.

12 The Bias Formula and Correction Procedure

Theorem 12.1 (Entropy Estimator Bias Formula)**.** *Let $\hat{\rho}$ be a maximum-likelihood or linear-inversion density matrix estimate from ν independent Pauli-basis shots. Then:*

$$\mathbb{E}[\hat{S}] = S_{\mathrm{vN}}(\rho) + \frac{d-1}{2\nu} + B_{\mathrm{SPAM}} + O\left(\frac{1}{\nu^2}\right) \tag{B3.1}$$

where $d = 2^n$, ν is the shot count, and B_{SPAM} is the SPAM (State Preparation and Measurement) bias calibrated independently.

Derivation sketch. The von Neumann entropy $S_{\mathrm{vN}} = -\sum_j \lambda_j \log_2 \lambda_j$ is concave in the eigenvalues λ_j. Each eigenvalue fluctuates as $\hat{\lambda}_j = \lambda_j + \xi_j$ with shot-noise variance $\sim \lambda_j(1-\lambda_j)/\nu$. By Jensen's inequality for concave functions, $\mathbb{E}[\hat{S}] \geq S_{\mathrm{vN}}(\rho)$. The leading-order gap uses $\lambda_j \approx 1/d$ (valid in the plateau regime): Bias $= \sum_j \frac{1}{2\nu \ln 2} \cdot \frac{\mathrm{Var}[\hat{\lambda}_j]}{\lambda_j} \approx \frac{d-1}{2\nu}$. The SPAM bias B_{SPAM} is an additive systematic from readout errors that is independent of ν and must be calibrated independently. □

12.1 Numerical Example: 15-Qubit GHZ State

For a GHZ state ($S_{\mathrm{vN}}^{\mathrm{phys}} = 0$) with $n = 15$, $d = 32768$, $\nu = 1000$ shots, and typical hardware $B_{\mathrm{SPAM}} \approx 0.05$ bits:

Table 4: Entropy plateau decomposition for a 15-qubit GHZ state.

Contribution	**Formula**	**Value (bits)**
True physical entropy	$S_{\mathrm{vN}}^{\mathrm{phys}} = 0$	0.00
Finite-shot bias	$(d-1)/(2\nu) = 32767/2000$	≈ 16.4
(as fraction of max = 15 bits)		$\approx 109\%$
Observed entropy (uncorrected)		> 15 bits

The finite-shot bias *exceeds the maximum possible entropy.* The maximum-likelihood estimator's positivity constraint ($\hat{\rho} \geq 0$) clips the unphysical range and produces a ceiling near the maximally mixed state — which is precisely the 40–50% plateau. This is not a hardware artifact; it is a mathematical consequence of fitting $\sim 10^9$ parameters from 3000 data points.

12.2 Correction Procedure

Given Eq. (B3.1), the corrected entropy estimate is:

$$\hat{S}^{\mathrm{corr}} = \hat{S}_{\mathrm{raw}} - \frac{d-1}{2\nu} - \hat{B}_{\mathrm{SPAM}} \tag{B3.2}$$

This correction requires only d (known) and ν (known) — it does not require knowing $\mathcal{F}_Q(\rho)$ and is therefore not circular. The uncertainty on the corrected estimate is approximately $\sigma[\hat{S}^{\mathrm{corr}}] \approx \sqrt{[\mathcal{F}_Q(\hat{\rho})]^{-1}/\nu}$, evaluated at the estimated (not true) state.

13 Falsification Criterion — Paper B

Paper B Falsification Criterion. The constraint-manifold explanation of the entropy plateau is falsified if:

1. On any platform where ν is scaled to match the $2^{n/2}$ threshold of Eq. (B2.4), the reported entropy does *not* converge toward the true physical entropy as predicted by (B3.1), *or*
2. The bias $(d-1)/(2\nu)$ from (B3.1) is not statistically consistent with calibration states of known entropy.

Either outcome would imply that the entropy plateau is physical decoherence, not a statistical artifact.

This experiment is runnable today on IBM Quantum, IonQ, or Quantinuum platforms. It requires only: (1) a calibration set of states with verified entanglement depth; (2) systematic variation of ν; (3) comparison of raw vs. corrected entropy to the prediction of Eq. (B3.1).

Paper C: Landauer's Principle in the ETI Framework

14 Introduction

Landauer's principle [9] states that logically irreversible computation (specifically, the erasure of one bit of information) must dissipate at least $k_{\mathrm{B}}T \ln 2$ of heat into the environment. This result has been stated in many forms since 1961, some rigorous [10] and some vague. The vague versions invite misinterpretation: treating information as a substance with ontological weight, claiming consciousness is thermodynamically special, or arguing that reversible computation is free in practice.

The ETI (Entropy-Thermodynamics-Information) framework gives Landauer's principle the same logical structure as the laws of thermodynamics: a small set of explicit axioms from which all results follow as theorems, with each axiom independently testable and each result traceable to the axioms on which it depends.

15 The Five ETI Axioms

The five axioms are ordered by decreasing certainty. A1 is among the most precisely verified claims in physics; A5 is an operational statement about resource constraints.

Axiom 1 (Causal Closure)**.** The universe $\mathcal{U}$ is a closed, causally connected system with no external agents, no external entropy reservoirs, and no information channels outside $\mathcal{U}$.

Physical content: Equivalent to global energy-momentum conservation and causal completeness. The basis of the second law: total entropy is non-decreasing because there is nowhere to export it.

Axiom 2 (Unitary Microdynamics)**.** Closed quantum systems evolve unitarily: $U(t) = e^{-iHt/\hbar}$. Open subsystems evolve through completely positive trace-preserving (CPTP) maps $\mathcal{E}$ on their density operators.

Physical content: Among the best-verified facts in physics. CPTP maps include all decoherence, thermalization, and measurement as special cases. Key consequence: $S_{\mathrm{vN}}(U\rho U^\dagger) = S_{\mathrm{vN}}(\rho)$ (unitary evolution preserves entropy; entropy increase in a subsystem requires export to the environment).

Axiom 3 (Effective Thermodynamics)**.** Thermodynamic entropy $S(\rho) = -k_{\mathrm{B}} \operatorname{Tr}(\rho \ln \rho)$ is a coarse-grained description relative to a chosen macroscopic partition. It is not a fundamental property of reality independent of description.

Physical content: Entropy is relational. The same physical state has different entropy values relative to different coarse-grainings. This prevents the reification of entropy and connects thermodynamic, information-theoretic, and quantum entropic notions under a unified framework.

Axiom 4 (Physical Memory)**.** Logical information is always instantiated in physical substrates satisfying: (i) *distinguishability* (distinct logical states correspond to non-overlapping supports in Hilbert space); (ii) *persistence* (metastability on computation-relevant timescales); (iii) *coherence stability* (no spontaneous decoherence faster than gate rate).

Physical content: Information is not free. A memory bit with no energy barrier between $|0\rangle$ and $|1\rangle$ is not a usable memory. The energy cost of memory maintenance (cooling, trapping, shielding) is distinct from but additive to the Landauer cost.

Axiom 5 (Finite Resources)**.** Any physical agent operates under finite memory capacity, finite cooling power, and finite control precision. Indefinite information storage without resource expenditure is impossible under A1–A4.

Physical content: Connects the framework to devices that actually exist. Rules out perpetual-motion computing. Implies a minimum power consumption for any computation lasting time t with n active components.

16 Landauer's Principle as a Theorem

Theorem 16.1 (Landauer's Principle from ETI Axioms)**.** *Under axioms A1–A5, any logically irreversible operation that reduces the logical state space of a memory register from k distinguishable states to k' states ($k' < k$) requires dissipating at least:*

$$Q_{\min} \geq k_{\mathrm{B}} T \ln\left(\frac{k}{k'}\right) \geq k_{\mathrm{B}} T \ln 2 \tag{C1.1}$$

of heat into the environment, where T is the temperature of the heat reservoir and the lower bound is achieved for the erasure of one bit ($k = 2$, $k' = 1$).

Proof sketch. By A2, the global (system + environment) evolution is unitary, so total entropy is conserved. By A3, the local (system) entropy after erasure is lower than before (fewer distinguishable states $\rightarrow$ lower entropy). By A1, the missing entropy cannot leave the universe; by A2 it cannot be destroyed; therefore it must be transferred to the environment (heat bath). The minimum transfer consistent with thermodynamics (A3) is $k_{\mathrm{B}} T \ln(k/k')$ by Boltzmann's formula. QED. □

16.1 What This Does Not Claim

- It does not claim that physical implementation of reversible computation has zero cost. A4 and A5 together imply that maintaining the physical substrate of a memory register requires continuous resource expenditure. The Landauer bound is the *minimum irreducible cost* of the logical erasure step only.
- It does not claim that consciousness is thermodynamically special. A1–A5 apply to any physical information-processing system.

- It does not claim that the bound is always achieved in practice. Achieving the bound requires quasi-static erasure in contact with a thermal bath — an idealization that approximates practical erasure only at very low speeds.

17 Falsifiable Predictions — Paper C

P1 (Landauer bound in quantum circuits). A quantum circuit performing m irreversible gate operations on n qubits at temperature T must dissipate at least $m\, k_{\mathrm{B}} T \ln 2$ of heat to its environment. Any measurement of dissipation below this bound falsifies A2 (non-unitary microdynamics) or A3 (entropy is not coarse-grained classical thermodynamics).

P2 (Erasure cost scaling). For a memory register at temperature T, the minimum erasure cost per bit is $k_{\mathrm{B}} T \ln 2$, independent of the physical implementation (electronic, photonic, atomic). Platform-dependence of the minimum bound would falsify A4 (that information is substrate-independent).

P3 (No perpetual computation). No finite physical system can perform unbounded computation (growing entropy, sustained indefinitely) without continuous energy input. Claimed violations falsify A1 or A5.

All three predictions are testable with existing laboratory equipment. P1 and P2 have been tested indirectly through Landauer experiments in classical nanoscale systems [13]; direct quantum circuit tests remain incomplete.

Synthesis: The PEIG Framework Connection

18 Physical PEIG: An Integrating Narrative

The three papers above are independently submittable and independently falsifiable. They are connected, however, by a common mathematical structure that organizes the dynamical evolution of quantum systems through a four-phase sequence. We describe this structure here for completeness, with careful labeling of its epistemic status.

18.1 The Physical PEIG State Vector [ESTABLISHED for P, E, I] [HYPOTHESIS for G]

For an n-qubit system, the PEIG state at time t is:

$$\mathrm{PEIG}(t) = (P(t),\ E(t),\ I(t),\ G(t)) \tag{S1.1}$$

P — Potential (Constraint Manifold Volume):

$$P(t) = \log_2 \dim \mathcal{C}_{\mathrm{obs}}(r_t) = \log_2(4^n - \mathrm{rank}(C_t) - 1) \tag{S1.2}$$

P measures the logarithmic size of the constraint manifold given the measurement history at time t. It decreases monotonically as measurements accumulate (Paper B provides the formalism). Range: $0 \leq P \leq 2n$ bits.

E — Energy (Gradient Flow Rate):

$$E(t) = \left\| \frac{d\rho}{dt} \right\|_F = \left\| -\frac{i}{\hbar}[H, \rho] + \mathcal{L}_D(\rho) \right\|_F \tag{S1.3}$$

The Frobenius norm of the state's rate of change, combining Hamiltonian evolution and Lindblad dissipation. E measures how rapidly the system traverses the constraint manifold.

I — Identity (Attractor Proximity):

$$I(t) = 1 - D_{\mathrm{HS}}(\rho(t),\ \rho_\infty), \quad D_{\mathrm{HS}}(\rho, \sigma) = \frac{\sqrt{\mathrm{Tr}[(\rho - \sigma)^2]}}{\sqrt{2}} \tag{S1.4}$$

ρ_∞ is the stationary state (Lindblad attractor). $I = 0$: maximally far from attractor. $I = 1$: at attractor. The Landauer cost of the evolution (Paper C) is paid as the system moves from $I \approx 0$ to $I \approx 1$.

G — Geometry (Entanglement-Sourced Curvature):

$$G(t) = \frac{\tilde{\kappa}}{k_{\mathrm{B}} \ln 2}\, S_{\mathrm{ent}}(t) \tag{S1.5}$$

The gravitational source term from Paper A. G starts large and negative (repulsive: high entanglement entropy, high P, low I) and evolves toward zero as decoherence reduces S_{ent}. The G component is labeled [HYPOTHESIS] because it depends on the Paper A extrapolation.

18.2 Four-Phase Physical Interpretation

Table 5: The four PEIG phases in physical terms.

Phase	**State**	S_{ent}	I	**Gravity (hypothetical)**
P (Potential)	High superposition	High	≈ 0	Repulsive
E (Energy)	Rapid Lindblad evolution	Decreasing	Rising	Transitioning
I (Identity)	Near Lindblad attractor	Near minimum	≈ 1	Minimal
G (Geometry)	Stable, structured	Fixed low value	Fixed ≈ 1	Weak attractive

18.3 The Epistemic Boundary — Physical vs. Cognitive PEIG

The PEIG formalism has been applied by analogy to cognitive systems (organizations, minds, AI) using P = option-space entropy, E = cognitive throughput, I = stable identity coherence, G = influence on surrounding systems.

The Cognitive PEIG is an analogy, not a derivation.

A human brain does not literally curve spacetime via the modified Einstein equation. The cognitive G is a structural metaphor for influence; the physical G is the Einstein tensor. They are mathematically analogous; they are not the same physical quantity. Publishing the cognitive PEIG as physics would be a category error. The cognitive analogy is scientifically suggestive as a framework for studying four-phase dynamics in complex systems. It is not a physics claim.

18.4 Open Problem OP5

The full coupled PEIG ODE system has not been written. Only $dI/dt = \alpha E - \beta I$ exists in formal derivation. The functions dP/dt, dE/dt, dG/dt are narratively described but not derived from first principles. Physical PEIG cannot be quantitatively falsified as a complete dynamical system until these equations are written and analyzed. This is an open problem for future work, not a blocking issue for Papers A–C.

19 Discussion

19.1 What Has Been Accomplished

This program has produced three independent scientific contributions at different stages of maturity:

Paper B is the most immediately useful and most securely established. The entropy plateau result has a clean derivation, a numerically concrete bias formula, a correction procedure that works today on existing hardware, and a falsification criterion that is testable with existing resources. It addresses a real and widespread misinterpretation in the NISQ tomography literature.

Paper C provides axiomatic clarity for Landauer's principle that advances over the existing literature by making the logical structure of the derivation explicit and traceable. The five axioms are individually testable, and the derivation is clean.

Paper A is the most speculative and the most significant in potential impact. The modified field equation is dimensional consistent; the coupling estimate is internally coherent under an explicitly stated hypothesis; the falsification criterion is carefully designed. The open problems (OP3, OP10) are real limitations that are honestly documented rather than hidden. The paper is a legitimate scientific contribution because: it is precisely stated, it is falsifiable, it is honestly bounded, and it would be among the most significant results in gravitational physics if confirmed.

19.2 Anticipated Reviewer Concerns and Responses

gr-qc reviewer: "The extrapolation from horizon entropy to laboratory entanglement is unjustified." Response: Agreed. Section 4 states this explicitly. The paper's contribution is the falsifiable framework and the coupling estimate, not a derivation that Jacobson's result extends to non-horizon settings.

quant-ph reviewer: "The signal magnitude is indeterminate. How can this be tested?" Response: The falsification criterion (Sec. 7) is stated in terms of the measurable quantity $|\tilde{\kappa}|$ via bounds, not in terms of a predicted signal. Null results at F1–F5 sensitivity constrain $|\tilde{\kappa}|$ regardless of whether the signal is currently computable.

Mathematically conservative reviewer: "OP8 (Bianchi identity compatibility) is unresolved." Response: Correct, and labeled as such. The paper does not claim full consistency with general relativity; it presents a phenomenological framework and identifies what needs to be verified.

Experimentally skeptical reviewer (Paper B): "The bias formula assumes the estimator is near the maximally mixed state." Response: The approximation $\lambda_j \approx 1/d$ is valid precisely in the plateau regime — which is exactly the regime being explained. For states further from maximum entropy, the actual bias is smaller, making the formula a conservative upper bound.

19.3 Relation to Existing Literature

The modified Einstein equation in Paper A sits in a tradition that includes Jacobson (1995) [1], Verlinde (2017) [2], and the ER=EPR program [3]. The novelty is not the idea that entanglement and gravity are related (this is well-established at extreme scales) but the attempt to frame that relationship as a falsifiable laboratory hypothesis with a specific coupling estimate and experimental protocol.

Paper B connects to the quantum tomography literature on shadow tomography [12] and to the quantum Fisher information literature [7]. The specific result — that the entropy plateau has a bias formula $(d-1)/(2\nu)$ with a $2^{n/2}$ shot-scaling threshold — is, to the authors' knowledge, not previously stated in this explicit form in the NISQ context.

Paper C contributes to the Landauer principle literature [9, 10, 11] by providing an axiomatic structure that is more transparent than existing treatments and more explicitly observer-independent.

20 Conclusion

We have presented three connected results in quantum information and gravitational physics. The common thread is the relationship between information, entropy, and physical geometry.

Paper B provides an immediately actionable result for the quantum computing community: the entropy plateau in NISQ tomography is a statistical artifact removable by the formula (B3.2). The shot-scaling law $\nu \propto 2^{n/2}$ gives experimenters a concrete target for their measurement budgets.

Paper C provides a rigorous foundation for Landauer's principle that is both more transparent and more general than existing statements.

Paper A opens a question rather than closing one. Whether laboratory-scale entanglement entropy curves spacetime is, as of this writing, unknown. The framework presented here is one way of asking that question precisely enough to test it. The honest statement of what is and is not derived, and the five-condition falsification criterion, represent the scientific value of this contribution regardless of whether the coupling exists.

The most important result of this program may not be any of these three papers individually, but the demonstration that honest, rigorous, falsifiable theoretical physics can be done by an independent researcher without institutional resources — as long as the distinction between what is established and what is hypothesized is never compromised.

Appendix A: Notation and Constants

Table 6: Unified notation table for all three papers.

Symbol	**Meaning**	**Units / Range**
$G_{\mu\nu}$	Einstein tensor	m^{-2}
$T_{\mu\nu}$	Stress-energy tensor	$\mathrm{kg\,m^{-1}s^{-2}}$
$g_{\mu\nu}$	Metric tensor	dimensionless
G	Newton's constant	$\mathrm{m^3kg^{-1}s^{-2}}$
c	Speed of light	$\mathrm{m\,s^{-1}}$
$\hbar$	Reduced Planck constant	$\mathrm{J\,s}$
k_{B}	Boltzmann constant	$\mathrm{J\,K^{-1}}$
$\ell_P = \sqrt{\hbar G/c^3}$	Planck length	1.616×10^{-35} m
$\tilde{\kappa}$	Entanglement-gravity coupling	dimensionless
$\tilde{\kappa}_{\mathrm{eff}}$	Effective (screened) coupling	dimensionless
α_{screen}	Environmental screening factor	$[0, 1]$
S_{ent}	Entanglement entropy density	$\mathrm{bits\,m^{-3}}$
$S_{\mathrm{vN}}(\rho)$	Von Neumann entropy of state ρ	bits
$\mathcal{H}$	Hilbert space	$(\mathbb{C}^2)^{\otimes n}$
$\mathcal{D}$	Space of density operators	dim $4^n - 1$
$\mathcal{C}_{\mathrm{constr}}$	Constraint manifold	submanifold of $\mathcal{D}$
$\mathcal{C}_{\mathrm{obs}}(r)$	Observable subspace given history r	shrinks with k
$\mathcal{F}_Q(\rho)$	Quantum Fisher information	state functional
B_{SPAM}	SPAM bias in tomography	bits
n	Number of qubits	positive integer
$d = 2^n$	Hilbert space dimension	≥ 2
ν	Number of measurement shots	positive integer

Appendix B: Mathematical Corrections Applied

The following corrections were applied to source documents during preparation of this manuscript. Each is labeled with a correction code.

Table 7: Summary of mathematical corrections (MA1–MA8).

Code	Issue	Action
MA1	G_{eff} formula dimensionally invalid	Formula removed from all papers
MA2	$\tilde{\kappa} = -1/4$ presented as derivation	Relabeled "estimate under hypothesis"
MA3	S_{ent} used inconsistently as total/density	Standardized to "always a density"
MA4	$\Delta a(R) \propto 1/R$ used without justification	Replaced with $1/R^2$ from enclosed-source form
MA5	$\mathcal{F}_Q$ notated as $\mathcal{I}_Q^{-1}(S_{vN})$ (wrong)	Corrected to $\mathcal{F}_Q(\rho)$ (functional of state)
MA6	Rank condition for constraints not stated	Linear independence condition added explicitly
MA7	Negentropy substitution inverts all predictions	Removed; sign convention standardized
MA8	Verlinde year 2025 (incorrect)	Corrected to 2017 throughout

Appendix C: Open Problems Registry

Table 8: Open problems registry as of March 2026.

ID	Description	Severity	Status
OP3	α_{screen} derivation from Lindblad dynamics	Critical	Open
OP5	PEIG coupled ODE system	Medium	Open (non-blocking)
OP8	Bianchi identity compatibility	High	Open
OP10	Planck-scale renormalization of $\tilde{\kappa}$	Critical (long)	Open
OP11	MICROSCOPE $\to \tilde{\kappa}$ bound mapping	High	Resolved: bound removed

OP4 (radial scaling), OP6 (Verlinde year), and OP7 (Kasevich citation) have been closed. OP11 was resolved by removing the MICROSCOPE bound from Table 8: because both MICROSCOPE test masses are classical solids with $S_{\text{ent}} \approx 0$, the experiment does not constrain $\tilde{\kappa}$ without a separate argument that has not been supplied.

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arXiv Submission Notes

Five Title Options (Ranked)

1. **Gravitational Coupling to Entanglement Entropy Density: A Phenomenological Framework with Falsifiable Predictions** (Paper A standalone, arXiv gr-qc — strongest, most targeted)
2. **Entanglement, Geometry, and the Limits of Quantum Observability: Three Results in Quantum Information and Gravitational Physics** (This unified paper — appropriate for repository preprint)
3. **The Entropy Plateau in NISQ Tomography: A Constraint Manifold Analysis and Bias Correction** (Paper B standalone, arXiv quant-ph — immediately useful, lowest review barrier)
4. **Landauer's Principle from Five Axioms: An Observer-Independent Formulation in the ETI Framework** (Paper C standalone, arXiv quant-ph — clean axiomatic paper)
5. **Modified Einstein Equations from Entanglement Entropy: Coupling Estimate, Screening Analysis, and NISQ Falsification** (Paper A alternative title — more explicit about content)

Submission Strategy Recommendation

Submit Paper B first (arXiv: quant-ph, target: Physical Review A or npj Quantum Information). No blocking issues; independent of Papers A and C; immediately useful to the NISQ community; lowest review barrier. Establishes credibility for the program.

Submit Paper C second (arXiv: quant-ph, target: Foundations of Physics). No blocking issues; clean axiomatic structure; independent result.

Submit Paper A third (arXiv: gr-qc, cross-list: quant-ph). Requires: (1) Section 5 reformulated around NISQ target (complete); (2) signal magnitude stated as indeterminate (complete); (3) MICROSCOPE row removed (complete); (4) Bose 2017 vs. 2023 citation resolved before submission.

Author Note for Independent Researcher

This research was conducted independently, without institutional affiliation or grant support. The mathematical corrections and open problem registry represent a deliberate effort at scientific self-auditing prior to submission. The author welcomes correspondence regarding any of the three frameworks described herein and is actively seeking collaborators with expertise in atom interferometry, NISQ hardware tomography, and semiclassical gravity.